# Abdelrahman Khalifa

+20 114 250 5042 | Cairo, Egypt | abdelrahman.khalifa.gad@gmail.com | linkedin.com/in/abd002 | github.com/Abd002

## Education

**Minia University** — Minia, Egypt
*Bachelor of Science (B.Sc.), Computer and Systems Engineering* — *2020 – 2025*

- Grade: Very Good with Honors.
- Activities: Codeforces Expert; 4-time ECPC finalist (2021–2024); Treasurer, IEEE Student Branch.

## Experience

**Embedded Software Engineer** — Aug 2025 – Present
*EKSON* — *Egypt*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS for more deterministic timing, devicetree-based drivers and easier porting to other boards.
- Implemented closed-loop PID control of the pneumatic valves, with feedback from an MPU6050 IMU over I2C (calibration and a Mahony filter for orientation), plus safety stops and a watchdog.
- Developed the PyQt5 desktop app that controls the simulator over a UART protocol; cut its CPU usage and latency by about 40% (measured) using asyncio to wait for ESP32 and server messages concurrently, off the UI thread.
- Engineered remote operation and firmware updates: a Node.js/Express + MongoDB backend that the desktop app polls over REST, then flashes new firmware with esptool; GitHub Actions CI.

**Google Summer of Code Contributor** — May 2026 – Oct 2026
*The Linux Foundation (OpenPrinting)* — *Remote*

- Built the **Printers page for COSMIC Settings** (printer status, supply levels, print queues, adding printers) as 5 Rust crates, with the CUPS/IPP/DNS-SD backend isolated behind a varlink IPC service (cosmic-settings #2182, in review).
- Wrote Rust-to-C FFI bindings (libcups3 DNS-SD) and fixed crashes in libcups' Avahi code; 7 PRs accepted upstream.

**Embedded Linux Systems Intern** — Dec 2024 – Jan 2025
*Orange Digital Center (ODC)* — *Egypt*

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing. Scored 100% in the final evaluation.

## Projects

**AUTOSAR-Based V2X Vehicle** | *graduation project, STM32* — GitHub

- Implemented the firmware update path (an ESP32/RPi gateway sends the image over UART) and a dual-bank bootloader with CRC check and rollback.

**Seat Heater Control System** | *Tiva C, FreeRTOS* — GitHub

- Real-time temperature and heater control with UART diagnostics, using tasks, mutexes and event flags.

**IoT Device Communication System** | *C++, Yocto* — GitHub

- TCP and UDP multicast between a Pi and QEMU clients.

**Kernel Monitor** | *C, Linux, ARM, QEMU* — GitHub

- Character device driver that walks the process list under RCU and exports per-task IDs, memory, threads, context switches and CPU time to user space; run on QEMU.

## Technical Skills

**Languages**: C, Modern C++, Rust, Python (asyncio, PyQt5), Java, Bash, Verilog
**RTOS & firmware**: Zephyr (devicetree, west), FreeRTOS, ESP-IDF, OSEK/VDX (Trampoline), 32-bit ARM Cortex-M (STM32, TM4C), ESP32-S3, AVR, AUTOSAR, bootloaders with image verification (CRC/signature), FOTA, PID control
**Embedded Linux**: Linux, Yocto, Buildroot, kernel modules, device drivers, device trees, QEMU, Raspberry Pi
**Protocols & tools**: UART, I2C, SPI, CAN, TCP/UDP, REST; Git, GitHub Actions CI, CMake, Make, debugging
**Testing**: unit and integration (end-to-end) testing, pytest, Python scripting for build and test automation in CI